"Dollar" classes

Anonymous inner classes and their $ names

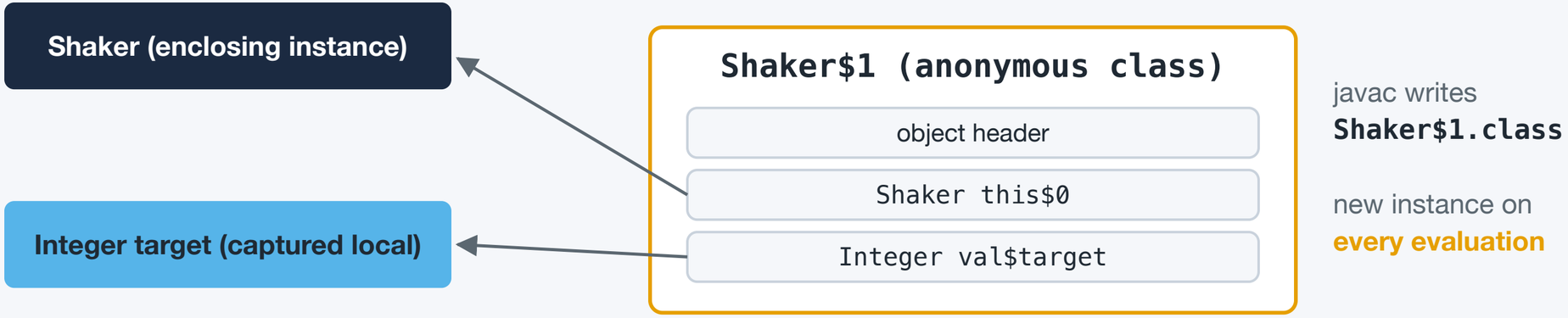
01

What is a "dollar" class?

An anonymous class is declared and created in one expression, without a name.

- `javac` still names it: `Outer$1`, `Outer$2`, ...
- Each one gets its own `Outer$1.class` file.
- Named inner and local classes use `$` names too.

What an anonymous class holds



- A new instance on every evaluation, even when nothing is captured.
- In an instance method it always holds the enclosing `this` (`this$0`).

In code

```
public class Shaker
{
    boolean hasMatch(List<Integer> list, Integer target)
    {
        return list.stream().anyMatch(new Predicate<Integer>()  // Shaker$1
        {
            @Override
            public boolean test(Integer each)
            {
                return each.equals(target);
            }
        });
    }
}
```

- Compare with lambdas (06): a non-capturing lambda is created only once.

References

- The Java Tutorials: Anonymous Classes
- JLS SE 25, 15.9.5: Anonymous Class Declarations
- JLS SE 25, 13.1: The Form of a Binary ($ names)
- JLS SE 25, 8.1.3: Inner Classes and Enclosing Instances